## Experiment 8: Multi-Agent Task Allocation and Coordination

### Aim

To implement a simple multi-agent task allocation system using Partial Global Planning (PGP) and Contract Net Protocol (CNP).

### Exercises

1. **Agent Setup**
   Create a Python-based system with 1 Manager Agent and at least 3 Delivery Agents. Add at least 6 delivery tasks with different costs/times for each agent.
2. **PGP Coordination**
   Create local plans for the agents. Create a conflict where two agents use the same location/resource at the same time. Detect the conflict and modify one of the plans to resolve it.
3. **CNP Task Allocation**
   Implement the CNP process:
   Task Announcement -> Bidding -> Bid Comparison -> Agent Selection -> Task Assignment.
4. **Multiple Task Allocation**
   Allocate all tasks automatically using the agents' bids. Display the final allocation as:

   Task | Selected Agent | Cost/Time

   Also calculate the total cost/time.

5. **Modification & Visualization**
   Change some task/agent costs and run the system again. Compare the results and plot Agent vs Total Assigned Cost/Time using Matplotlib.

### Observation

Observe how PGP resolves conflicts, how CNP assigns tasks, and how changing the input affects the final task allocation.